Exercise2 Aggregate & Operators

① Select Distinct department
From Student;

Department
IT
HR
Finance

② Select department,
AVG(age) AS age
From students
Group by department;

Department	Avg-age
IT	20.5
HR	22.0
Finance	23.0

③ Select department, Count(*) AS student_count
From students
Group by department
Having Count(*) > 1;

department	Student_count
IT	2
HR	2

④ Seled- student_id,
Name,
Age,
department
From Students
Where age BETween 21 AND 23;

student_id	Name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

⑤ Seled student_id;
Name,
age,
department
Where department IN('IT', 'HR')
AND age > 21

student_id	Name	age	department
2	Bob	22	HR
5	Eve	22	HR

⑥ Seled- department, Sum(credits) AS total credits
From courses
Group by department
~~Having~~ Having Sum(credits) > 5;

department	total_credits
IT	11

⑦ Select course_id,
Course_name,
department,
credits;
From Courses
Where credits <> 4;

course_id	course name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	statistics	HR	3

⑧ Select course_id,
Course name,
Credits
From Courses
~~Where~~ Order By credits Desc
limit 3;

Course_id	Course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

(9) Select
Max(grade) AS max_grade,
Min(grade) AS min_grade,
Avg(grade AS avg_grade
From Enrollments;

Max_grade (Highest grade)	Min_grade (Lowest grade)	Avg_grade (Average grade)

(10) Select course_id, Count(*) As enrollment_count
From Enrollments
Group by course_id;

Course_id	Enrollment_count
101	1
102	1
103	1
104	1
105	1

(11) Select
department,
Sum(Salary) As total_Salary,
Sum(bonus) As total_bonus
From Salaries
Group by department;

department	total_Salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

(12) Select department,
AVG(salary) AS avg_salary
From Salaries
Group By department
Having AVG(salary) > 55000;

department	avg_salary
Finance	70000
IT	61000

(13)

(14) Select department,
Sum budget AS total_budget
AVG budget AS Avg_budget
from projects
Group By departments
Having AVG(budget) > 70000

department	total_budget	Avg_budget
Finance	80000	80000
IT	27000	135000